

Score: 100%
13/13 points

WISH - Digital - Day-4

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Q1. What is the primary purpose of Design for Testability (DFT)?

- | | |
|--|---|
| ☐ a) Improve circuit performance | ☒ c) Make circuits easier to test |
| ☐ b) Reduce power consumption | ☐ d) Reduce manufacturing cost |

1/1 point

Q2. Which fault model assumes a node is permanently at logic '1' or '0'?

- | | |
|---|--------------------------------------|
| ☐ a) Bridging fault | ☐ c) Delay fault |
| ☒ b) Stuck-at fault | ☐ d) Open fault |

1/1 point

Q3. Which of the following is a common DFT technique?

- ☐ a) Clock gating
- ☐ c) Pipelining
- ☒ **b) Scan design**
- ☐ d) Load balancing

1/1 point

Q4. What is the main function of scan chains in DFT?

- ☐ a) Reduce power consumption
- ☐ c) Improve circuit speed
- ☒ **b) Enable easy test pattern application**
- ☐ d) Reduce transistor count

1/1 point

Q5. What is ATPG used for in DFT?

- ☒ **a) Generating test patterns automatically**
- ☐ c) Reducing chip size
- ☐ b) Increasing chip performance
- ☐ d) Improving power efficiency

1/1 point

Q6. Which DFT method allows testing without external test equipment?

- ☐ a) JTAG
- ☒ **c) Built-In Self-Test (BIST)**
- ☐ b) Scan chains

☐ d) Functional testing

1/1 point

Q7. Which of the following is NOT a structural testing method?

☐ a) Scan-based testing

☐ c) BIST

☐ d) Boundary scan

☒ **b) Functional testing**

1/1 point

Q8. What is the primary benefit of test compression?

☐ a) Increases test time

☐ c) Improves chip performance

☒ **b) Reduces test pattern volume**

☐ d) Reduces manufacturing cost

1/1 point

Q9. Which of the following is NOT an advantage of scan testing?

☐ a) Easy fault detection

☐ c) Requires extra hardware (scan flip-flops)

☒ **b) Reduces test time significantly**

☐ d) Allows automatic test pattern generation

1/1 point

Q10. What does MBIST stand for?

- ☐ a) Modular Built-In Self-Test
- ☐ b) **Memory Built-In Self-Test**
- ☐ c) Multi-Block Integrated Scan Testing
- ☐ d) Multi-Bit Integrated System Testing

1/1 point

Q11. In scan design, what type of flip-flops are used?

- ☐ a) Latches
- ☐ b) Edge-triggered flip-flops
- ☐ c) **Multiplexed flip-flops**
- ☐ d) JK flip-flops

1/1 point

Q12. Which test approach helps detect timing-related faults?

- ☐ a) Scan testing
- ☐ b) Functional testing
- ☐ c) **Delay testing**
- ☐ d) Bridging fault testing

1/1 point

13. What is the major drawback of DFT techniques?

☐ a) **Increased area overhead**

☐ c) Faster circuit operation

☐ b) Reduced power consumption

☐ d) Lower cost of manufacturing

1/1 point

Done



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